Group Project

Introduction

Transportation is a crucial topic in urban planning.

The goal of this group assignment is to devise a research question that can be answered by analysing a transportation dataset, that is, the origin-destination (OD) data of public transportation in Singapore (explained later), and to carry out a data analysis in form that is similar to a short research paper.

In this project, you will be using the OD dataset including bus and MRT trips, released by the LTA at their DataMall. The dataset is quite interesting, it contains detailed information about transportation patterns at a high spatial and temporal level of detail. Considering that the dataset is of geographic nature (bus stops and train stations have locations, and people travel from one point to another), it may be used also for a variety of spatial analyses.

To practice your data joining skills and to have this assignment transcend more than one domain, <u>it is mandatory that the transportation data is linked to at least one additional dataset not listed in this assignment</u>.

The group project is conducted in groups of up to six students. Conceptually, it is similar to the assignments so far: you have a dataset and you are free to do any analysis on your own.

Data

OD data is one of the key datasets in transportation, and it has been used in myriads of studies. It reveals transportation patterns: passenger volumes of stations at different times, and traffic of routes (i.e. volume between two stations).

The dataset can be downloaded through an API that is listed from the LTA DataMall. The same website contains also detailed documentation about the API and the dataset (API User Guide & Documentation). The dataset is actually a collection of two instances, which are organised by bus and MRT trips. They are called officially:

- Passenger volume by origin destination bus stops
- Passenger volume by origin destination train stations

The API provides this data at the monthly basis, for each combination of stop/stations, at different types of days (by weekdays and weekends), and at hourly intervals. While it does not include individual trips (hello privacy), it is rather aggregated. However, it is still quite detailed, enabling you to conduct a variety of interesting analyses.

An accompanying set of datasets is the data on *tap in and tap out passenger volume for individual bus stops and train station*, which is quite related to the datasets above but you may find them relevant as well and you may want to use one of these instead of the ones listed above. On the LTA DataMall, they are available as:

- Passenger volume by bus stops
- Passenger volume by train stations.

For your convenience, and since we did not spend much time learning how to work with APIs, I downloaded all the data above since January 2020 (both bus and MRT). The files are made available in the same week when the group project is announced during the lecture. The dataset may be updated during later weeks (i.e. LTA releases the data around the 15th day of the month for the previous month), so please monitor the provided link.

You may focus on any time period as you wish and as required by the analysis you intend to conduct.

If you are interested in learning more about APIs and how to fetch data using them, you are advised to download this dataset on your own. LTA DataMall is a well documented API that is fairly easy to use, so it presents a good learning opportunity for API beginners. This is optional.

Furthermore, this API and the data are well documented in a PDF that is available on the LTA Data Mall website (LTA DataMall API User Guide). I included the same PDF together with the files. Please refer to this PDF should you have questions about the data structure and content.

Do note that these datasets do not contain the geographic location (coordinates) of each bus stop and MRT station. If you will venture into carrying out a spatial analysis, as part of the assignment, you will have to make some effort in processing and linking the corresponding geographic data with the datasets above. To facilitate this process, an additional dataset is made available: the dataset on bus stops, which was also sourced from the LTA DataMall. It contains detailed information for all bus stops currently being serviced by buses, including coordinates. The dataset is given as collection of JSON files, for every 500 stops (as returned by the API). If you will use this dataset, you are expected to understand how to load it. As bus stops change over time, the data is given from multiple periods (e.g. some bus stops did not exist a year ago, while some others have been removed in the meantime).

Task

Your tasks are:

- Do a literature review to understand the state of the art of this topic, applications of this data, research gaps, and examples of analyses using similar data.
- Understand the provided dataset(s).
- Link the OD data with at least one or more additional datasets that is not listed in this assignment (e.g. amenities, population, real estate).
- Formulate a research question you want to answer.
- Answer the research question with a data analysis.

The deliverable of the project is a paper containing text and visuals (plots). The form of the report must be roughly similar to a short academic paper (containing an introduction, literature review, method, results, conclusion, references).

The scope that is expected:

- Up to 10 pages.
- At least 5 meaningful visuals (plots/maps/tables).
- In your analysis, include at least one additional dataset not listed in this project description.

You may get ideas what to do by exploring the data and papers that have been published on this topic. You are allowed to replicate existing research in Singapore or another location (i.e. do a similar analysis someone already did focusing on another city), as long as the original research is properly cited and understood. In that case, you can eventually do a comparison between the two studies. There are many things you can do with such data, especially if you link it to another urban dataset. For example, effect of opening of a new MRT line or bus stop on transportation volume and real estate.

Do note that while the scope of the Project actually does not seem substantially heavier than Assignment 2, and you have up to five times the resources (up to 5 group members instead of 1), it may require more effort in total, e.g. a brief literature review. Furthermore, unlike Assignment 2, which is released concurrently, the outcome of the Project is expected to be more *academic*: a (short) research paper. However, individual effort should not be more than what you invested in Assignment 2, and consequently the weight of the project in the mark is in about the same range as the weight of individual assignments.

Notes:

- You may focus on any time period you wish, you don't have to analyse all the months that are available in the data.
- You may focus on either bus or train trips, you do not necessarily have to focus on both.
- You do not have to use all the datasets above. Multiple datasets are given, but you do not have to use all of them if they are not relevant for the study you are intending to conduct.
- You may replicate an existing analysis, as long as it is properly cited and understood. If you opt for this path, do consider to provide the comparison in the results with respect to the original study and lessons learned.

READINGS ON HOW TO WRITE AN ACADEMIC PAPER

NUS is a research intensive university. Thus, it is important that our graduates are well versed with formulating a research question and are capable to write a scientific paper. Do spend some time and effort on these matters. For example, there are guides such as: https://writingcenter.gmu.edu/guides/how-to-write-a-research-question
and
https://www.washingtonpost.com/blogs/answer-sheet/post/a-guide-to-writing-an-academic-paper/2012/01/18/gIQAjGCTCQ_blog.html

and
http://www.jessicacalarco.com/teaching-resources

Another potentially relevant guide: how to write a paper in R Markdown: https://libscie.github.io/rmarkdown-workshop/handout.html
(Though you don't have to use R Markdown for this assignment.)

Groups

This is a group project. Groups may have up to five (5) students. It is also acceptable to have groups that are smaller than 5 people.

Please send me the list of group members and a brief description of the project (title, research question, goals, etc.) by email to me (filip@nus.edu.sg) by the stipulated deadline. **Please include the TA and all group members in Cc in your email.**

If you are without a group, please approach me after the deadline, and I will group the remaining students together or assign them to existing groups.

Process

The project has multiple stages:

FORMING GROUPS AND IDEAS (WEEK 9)

At this stage, you have to form a group and have a general idea what you want to do. Write a title and a paragraph about the formulated research question and the topic you plan to explore. Remember to provide me with the list of people that form your group. Afterwards, if the project you had in mind does not turn out to be feasible, it is fine to change your mind; we can discuss an alternative research question and topic. This is not graded and has no influence on your mark. It is intended to have you start thinking about your project and potentially get some preliminary feedback from me.

GETTING FEEDBACK IN THE CLASS (WEEKS 11-12)

As you progress with the work, in weeks 11-12 you will have some preliminary results, and you are welcome to discuss them and get guidance in the class. During these weeks, we will have a somewhat shorter session, so we will have some time left to discuss your work before the submission. Please come prepared with questions. If the consultation sessions will not be possible during this period, we will organise an alternative time.

FINAL REPORT (WEEK 13)

Please submit your report by the end of Week 13.

Learning goals of this assignment

This project concludes our module. Thus, it is intended to demonstrate and consolidate the knowledge you gained during the entire course, and whether you are able to connect the dots across the different topics in the module, and do so in a collaborative manner. It also gauges whether the skills you attained are transferrable to other kinds of datasets and domains. Hence the topic of the assignment is transportation, which we haven't covered in the lectures and tutorials of the module.

In general, this project is intended that you can demonstrate the ability to do the following:

- Come up with an idea for analysis, research question, and demonstrate that you can carry it out.
- Ability to do some research on a topic to get ideas and understand the state of the art, e.g. reading one research paper analysing a similar dataset.
- Understand and handle a large dataset, and reduce it to the relevant aspects.
- Understanding statistical learning.
- Joining different datasets.
- Describe the analysis and results well in form of an academic paper.
- Optional: carry out a spatial analysis and deal with geographic data (GIS) in the R environment.

Note that in this project, you are not expected to do every single thing we learned in the module. You are free to entirely omit some topics (e.g. if you don't think that machine learning is useful for your project, no need to do it then).

Tips

- Since this is the only assignment that is not individual, consider techniques such as pair programming (https://en.wikipedia.org/wiki/Pair_programming).
- OD data is popular in research. You may search for analyses done by others to get ideas.
- Transportation as a topic is a pillar of urban planning, and it is intertwined with many other topics such as demographics and amenities. You are strongly encouraged to include some of these aspects in your work.

Checklist

Based on the reports from last years, I noticed some common shortcomings and points for improvement, so I prepared a list of things that you may want to ask yourself before the submission, to help you avoid issues that I have observed so far, and score a high mark:

- Did you structure the document well?
- Does your report have an introduction that makes it clear what it is about?
- Do you have a clear research/analysis question?
- Did you answer the question properly and supported the answer with data and visuals?
- Did you make sure that the plots and maps are well styled and annotated, and easy to understand?
- Did you describe the conclusions from the visuals, helping the reader to interpret them?
- Did you check the licence and quality of the data, and described sufficiently additional datasets that you may have used?
- Did you check the metadata of the datasets you used?
- Did you remove unnecessary warnings and messages that clutter the report in R Markdown?
- Is your report free of grammar error and written in clear English?
- Does your report have a clear conclusion?
- Did you refer to all external materials you used?
- Did you attribute properly all the data?
- Does your report contain sufficient text to lead the reader?
- Did you include a literature review, describing the state of the art and related work?
- Did you cite all sources?

Deadlines

Please submit the report by **Sunday 19 April by end of day.**

This date is given well in advance, to give you more than a plenty of time to work on the project.

Please note that it is not possible to submit the report with a delay, and there cannot be extensions due to the end of the semester. Here is the full list of deadlines:

Stage	Deliverable	Deadline	Channel
Forming the groups and project tasks	- Names of group members (up to five, but it's also possible to work in groups of fewer) - Title, and short description about what you want to do and the research question you want to answer	Wednesday 18 March by EOD	email
Final paper	- Detailed report about your project	Sunday 19 April by end of day [the end of Week 13]	Canvas

Submission of the paper

The final report has to be roughly in the form of an academic paper (containing Introduction, literature review, method, results, conclusion, references).

Please submit your report on Canvas, a folder has been created for the submissions.
Do not submit by email.

Unlike the individual assignments, for the group project, you do not have to use R Markdown in preparing the report. For example, you can write the report in Word and include visuals generated in R.

One submission per group is sufficient.

Please submit your report in PDF. The report must be submitted as a file, not as a link to an online resource.

Please submit a single file. If you must submit multiple files, please zip them.

When submitting the report please make sure that your report has a title, abstract, and that your names are somewhere at the top of the report next to the title. For the structure, please refer to various academic papers that are plentiful online.

If you want to include materials that do not fit within 10 pages, you may want to consider an appendix.

Marking

This project is worth 30% of your final mark. A project will get a single mark based on the report, and all students in a group will get the same mark. There will be no peer review, so make sure you all contribute more or less equally to the work.

Please note that this group project is not a zero-sum game in the class. This is no competition among groups, and all groups will be marked based on the quality of their work and presentation, and not comparing them to other groups. All groups can score a high mark if the work is of high quality.

You'll be graded based on your understanding of the topics in the module, quality of the report, creative thinking, criticism and honesty, ideas, depth of the analysis, and interpretation of the data and results.

You will get more points for:

- Thinking outside the box.
- Formulating an interesting research question.
- Going beyond the scope of the class, e.g. finding new functions and using plot types not covered in the class, i.e. demonstrating the capability to find new solutions.
- Coming up with interesting conclusions and producing interesting plots/maps.
- Producing an analysis with novel results, which has not been done before.
- Producing an academic paper. You are not expected to write a top notch paper to be submitted to a top tier academic journal, but rather structure-wise it has to look like one and has to be written in a clear and concise manner expected at the graduate level at NUS.

Questions?

If you have questions please use MS Teams (private message or group chat). I'll be happy to help within the weekdays and usual working hours, but please spend some time first on finding a solution on your own. It's perfectly fine to get stuck, and debugging is part of the learning experience.

If you have a personal matter feel free to get in touch by email (filip@nus.edu.sg). Please do not use email if you have technical questions, it'll be easier and more efficient for both of us to use Teams.

Rules and policy

This course operates under the NUS Code of Student Conduct.

Consultation with other groups is of course allowed and encouraged, but it must be limited to discussion, not common solving of assignments. Sharing assignments and other materials is a violation of academic honesty policies.

You may be aided by online resources (and this is encouraged, widening your knowledge, not to mention that most programmers spend half their time googling), and you may even copy code fragments from online (e.g. from StackExchange) as long as you understand what you are doing, that you do so to a reasonable extent, and that you acknowledge these sources.

Please note that, following best practices and crediting others, it is necessary to add attribution to any aspect (data, etc.) that was used to complete the work. You may want to check the requirements of data sources, used packages/plugins, etc. These are usually specified in the metadata and documentation.

Credits

- LTA for making the OD data available.
- Photo in the header by Shawn Ang on Unsplash

What's the next assignment about?

There isn't one, you are done. ✔

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Good luck, and have fun with the data.